A Measure-and-Prepare Framework for Arbitrary Quantum Gate Excision and Its Applications

Tian-Fu Chen[1,4,5], Yu-Hung Pan[2,4,5], Ting-Hsun Chi[3], Po-Hung Chen[3], Dah-Wei Chiou[2], and Jie-Hong R. Jiang[1,2,3,4,5]
[1]*Graduate School of Advanced Technology, National Taiwan University*, Taipei, Taiwan
[2]*Graduate Institute of Electronics Engineering, National Taiwan University*, Taipei, Taiwan
[3]*Department of Electrical Engineering, National Taiwan University*, Taipei, Taiwan
[4]*Center for Quantum Science and Engineering, National Taiwan University*, Taipei, Taiwan
[5]*Physics Division, National Center for Theoretical Sciences*, Taipei, Taiwan
{d11k42001, r12943105, b10901184, b10901076, dwchiou, jhjiang}@ntu.edu.tw

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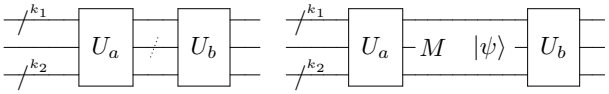
***Abstract*—The wire-cutting technique has been recently developed to partition a large quantum circuit into smaller subcircuits, enabling distributed computation across multiple quantum devices with limited capacity. In this work, we generalize the standard wire-cutting method—which can be interpreted as excising an identity operation—to a broader framework that allows the excision of arbitrary unitary gates. We propose a general measure-and-prepare approach to realize this extension, which allows a quantum gate to be excised from a circuit significantly more efficiently than by cutting all the wires surrounding the gate, leading to exponential savings in classical computational cost with respect to the number of wires involved. As an application, we employ this framework for the classical simulation of near-Clifford quantum circuits by excising non-Clifford gates from the circuit and efficiently simulating the remaining Clifford subcircuit in polynomial time. Experimental results demonstrate the superiority of our method over the prior wire-cutting technique in emulating near-Clifford circuits.**

***Index Terms*—Distributed quantum computing, wire cut, gate excision, Clifford circuits, quantum circuit simulation**

I. Introduction

Quantum computing has progressed rapidly beyond the reach of classical machines. However, quantum computers are still highly susceptible to errors due to noise, decoherence, and other technical challenges that limit their reliability and scalability. To distribute quantum computation over a set of capacity-limited quantum devices, the wire-cutting technique [1] is developed to partition a large quantum circuit into smaller subcircuits. The subcircuits are then independently executed with various pre-defined measurement and state preparation configurations, and the execution results are combined through classical postprocessing.

In this work, we propose a measure-and-prepare framework based on the wire-cutting technique. Unlike prior wire cutting, which leaves all gates unchanged, the proposed framework allows excising a quantum gate and implicitly simulating it through a weighted sum of measure-prepare configurations. Because the excised quantum gate is excluded from the circuit, the excision reduces the execution cost of the gate and is helpful in applications where some quantum gates are expensive to implement or simulate. For example, T-gates may have a quantum cost hundreds of times higher than Clifford gates in fault-tolerant quantum computation [2]. The proposed framework can help to exclude the T-gates from actual difficult execution or simulation.



(a) Circuit before wire cut. (b) Circuit after wire cut.

Fig. 1. Illustration of wire cut.

As an application, we implement a pseudo-quantum simulation framework, `PQSim`, for near-Clifford quantum circuit simulation, which adopts the proposed framework to excise non-Clifford gates, leaving a purely Clifford subcircuit that can be efficiently simulated. Experimental results demonstrate the superiority of our emulator in simulation speed compared to conventional wire-cutting techniques.

II. Preliminaries

We assume readers have basic knowledge of quantum computation, including pure states, mixed states, quantum gates, and measurement observables. In this section, we provide a brief introduction to the concept of the wire-cutting technique.

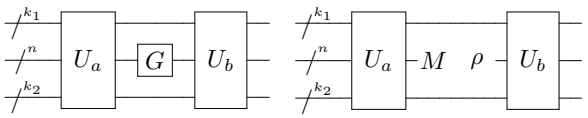
To execute or simulate a quantum circuit on a smaller device with limited qubit capacity, recent work [1] proposes a method to partition a quantum circuit into several subcircuits. Figure 1 demonstrates applying the method on a cut point. Due to the cut, an extra circuit input and an extra circuit output are produced at the cut point. The method requires iterating over 8 different measure-prepare configurations of the measurement observable M and the input state $|\psi\rangle$ at the cut point, including $(M, |\psi\rangle) \in \{(I, |0\rangle), (I, |1\rangle), (X, |+\rangle), (X, |-\rangle), (Y, |i\rangle), (Y, |-i\rangle), (Z, |0\rangle), (Z, |1\rangle)\}$. The other circuit inputs and outputs follow the original circuit specifications. We then calculate the expectation value under each measure-prepare configuration. The values of all configurations are collected and weightedly summed, and the result would equal the expectation value of the original circuit. We notice that this method can be recursively extended to any number of cuts, and the original circuit may be partitioned into many unconnected subcircuits that can be executed on capacity-limited devices after enough cut points are selected.

III. Measure-and-Prepare Framework for Arbitrary Quantum Gate Emulation

We generalize the idea in [1] and propose an extended measure-and-prepare framework that enables the simulation of arbitrary quantum gates in a similar manner. Using this

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(a) Circuit before gate excision. (b) Circuit after gate excision.

Fig. 2. Illustration of the measure-and-prepare framework.

framework, any quantum gate can be excised from a circuit and replaced with a corresponding measure-and-prepare process, thereby avoiding the need to implement certain high-cost quantum operations.

By straightforwardly generalizing the proof provided in [1], one can show that replacing an arbitrary n-qubit gate G with a weighted sum of measure-and-prepare channels—each specified by a tuple (c_i, O_i, ρ_i) of the associated weight c_i, measurement operator O_i, and preparation state ρ_i—preserves the value of the tensor network of the whole circuit if $GAG^\dagger = \sum_i c_i \cdot \mathrm{Tr}(AO_i) \cdot \rho_i$ holds for all $2^n \times 2^n$ matrix A. We first note that for any $2^n \times 2^n$ matrix A, we always have

$$A = \sum_{P \in \{I,X,Y,Z\}^{\otimes n}} \frac{1}{2^n} \cdot \mathrm{Tr}(AP) \cdot P. \tag{1}$$

For any n-qubit quantum gate G, it follows

$$GAG^\dagger = \sum_{P \in \{I,X,Y,Z\}^{\otimes n}} \frac{1}{2^n} \cdot \mathrm{Tr}(AP) \cdot GPG^\dagger. \tag{2}$$

Since $GPG^\dagger$ can always be expressed as a linear combination of at most 4^n Pauli matrices—as can be seen by applying Eq. (1) with $GPG^\dagger$ substituted for A—we have

$$GAG^\dagger = \sum_{P,P' \in \{I,X,Y,Z\}^{\otimes n}} w(P,P') \cdot \mathrm{Tr}(AP) \cdot P', \tag{3}$$

for weight coefficients $w(P,P') = \frac{1}{2^{2n}}\mathrm{Tr}(GPG^\dagger \cdot P') \in \mathbb{R}$ that depend on P and P'. Therefore, we can always find the weight coefficients $w(P,P')$ to make Eq. (3) hold.

Therefore, applying the n-qubit gate G can be transformed into iterating over the 16^n measure-prepare configurations, as illustrated in Figure 2, with $M = P$ being the measurement observable, $\rho = P'$ being the initial state, and $w(P,P')$ being the weight. Note that the transformation approach is systematic, automated, and applicable to arbitrary quantum gate. Moreover, the measure-prepare configurations with zero weights can be excluded from iterations, so the practical required number of iterations can be much lower than 16^n. As an example, excising a T-gate results in 6 measure-prepare configurations with non-zero weights, while excising a Toffoli gate results in 232 measure-prepare configurations. We note that these numbers are much lower than directly adopts the existing wire cut technique to detach a gate by cutting all its surrounding wires, which leads to 64 and 262144 measure-prepare configurations for a T-gate and a Toffoli gate, respectively.

IV. Applications to Near-Clifford Circuit Simulation

Clifford circuits are a special type of quantum circuits that contain only Clifford gates, which can be generated by the gates $\{H, S, \text{CNOT}\}$. The Gottesman–Knill Theorem [3] states that Clifford circuits can be classically simulated in polynomial time. However, even a single non-Clifford gate can ruin the efficient simulation feature of Clifford circuits, forcing us to resort to exponential-cost general quantum circuit simulation. To overcome this bottleneck, we can apply our proposed measure-and-prepare framework to excise non-Clifford gates from a near-Clifford circuit, resulting in a Clifford subcircuit to be efficiently simulated for a number of times, maximally utilizing the efficient simulation feature of Clifford circuits.

Incorporating with the *ancilla-assisted initialization* technique proposed in [4] and the *algebraic representation* technique that represents expectation values of measurement observables in integer tuples, our our method is hardware-friendly for the implementation on FPGAs, which benefits from the speedup of hardware emulation. Therefore, we implement our proposed method, `PQSim` (Pseudo Quantum Simulator), with the `Nexys Video Artix-7` FPGA accessed by a laptop equipped with AMD Ryzen 9 5900HX CPUs.

We conduct experiments on 40-qubit random circuits with 1 to 10 T-gates and compare our method with the vanilla version `PQSim_v` that directly adopts the existing wire cut technique to detach all non-Clifford gates. Our code and benchmarks are available at [5]. The reported runtime includes CPU-FPGA interfacing time but omits pre-processing time on circuit synthesis, FPGA burn-in, etc, which are one-time process as the burned-in FPGA can be used repeatedly. Experimental results showed that `PQSim` obtained the expectation values of the specified measurement observables in the 10-T circuit within 115 seconds. As a comparison, `PQSim` ran 5248 times faster than `PQSim_v` in the case of 4 T-gates, the largest case that `PQSim_v` finished within the 10-minute timeout limit. These results show the superiority of the proposed measure-and-prepare framework and hardware emulation in near-Clifford circuit simulation.

Acknowledgments

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[5] https://github.com/NTU-ALComLab/PQSim.

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